

ABE REVIEW

Q&A IN MATHEMATICS, STATISTICS AND BASIC ENGINEERING

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Easy Questions

1. Based on PhilMech's 2011 and 2012 agricultural statistics of Asian countries, the Philippine farm mechanization showed an upswing trend which recorded an average mechanization level of 1.23 hp/ha; rice and corn farms registered the highest level of available farm power at 2.31 hp/ha (Philippine Official Gazette, 2013). For the Philippines to level up with Japan, the leader in Asia, agricultural and biosystems engineers should know Japan's statistics. The level of farm mechanization in Japan as of 2012 was 7 hp/ha. What statistical approach was used to get this data?

A. Survey B. Sampling C. Telephone interview D. Simulation

Answer: A

2. Based on the agricultural statistics survey of PhilMech in 2011 and 2012, 70 percent of the total farm power is available for use in production operations while the remaining percentage is used for postharvest operations. What main statistical parameter was used?

A. Mode B. Mean C. Median D. Variance

Answer: B

3. The abstract science of number, quantity and space.

A. Numerology B. Quantitative math C. Mathematics D. Spatial science

Answer: C

4. Mathematics as it is applied to other disciplines such as physics and engineering.

- A. Pure math B. Advanced math C. Applied physics D. Applied math

Answer: D

5. Mathematics that is done without explicit or immediate consideration of direct application.

- A. Pure math B. Advanced math C. Applied physics D. Applied math

Answer: A

6. The equivalence of $\ln e^{(y+z)/2}$.

- A. $0.5(y+z)$ B. $(e^y + e^z)/2$ C. $\log 2 \times e^{(y+z)}$ D. $\ln [(y+z)/2]$

Answer: A

Solution:

From the rules on natural logarithm, $\ln e^x = x$

$$\begin{aligned}\ln e^{(y+z)/2} &= (x+y)/2 \\ &= 0.5(x+y)\end{aligned}$$

7. The factorial of 10 less factorial of 9.

- A. 3,628,800 B. 362,880 C. 3,265,902 D. 3,265,920

Answer: D

Solution:

$$\begin{aligned}10! - 9! &= 3,628,800 - 362,880 \\ &= 3,265,920\end{aligned}$$

8. In an experiment, the total degrees of freedom increase as _____ increases.

- A. Level of significance B. Replication D. Error D. Mean square

Answer: B

9. In a rice yield experiment, pre-experiment data shows varying soil fertility in the field. What experimental design should be used such that the effects of blocking can be determined?

A. RCBD

B. CRD

D. Latin square

D. DMRT

Answer: A

10. A ranch is composed of 2,500 heads of Holstein-Friesian and 7,500 heads of Brown Swiss cattle. Thirty representative heads are to be selected at random for use in an experiment. What is the probability that a Brown Swiss will be randomly picked up at first selection?

A. 25%

B. 50%

D. 75%

D. 100%

Answer: C

Solution:

$$\begin{aligned}\text{Probability} &= \text{Number of heads of Brown Swiss} / \text{Total Number of Heads} \times 100\% \\ &= 7,500 / (2,500 + 7,500) \times 100\% \\ &= 75\%\end{aligned}$$

Moderate Questions

11. A single-factor experiment using completely randomized experimental design results to the following statistical values: treatment sum of squares of 2,510.5, error mean square error of 13.4, error degrees of freedom of 12 and total degrees of freedom of 14. How many treatments are involved?

A. 3

B. 4

D. 5

D. 6

Answer: A

Solution:

$$\text{Tr DF} = \text{Total DF} - \text{Error DF} = 14 - 12 = 2$$

$$\text{No. of treatments} = \text{Tr DF} + 1 = 2 + 1 = 3$$

12. In random sampling for the location of different treatments in the field, the labels for Treatments 1, 2, 3, 4 and 5 were placed inside a box. Treatment 3 came out in the first pick. What is the probability that you can pick Treatment 1 in the second pick?

A. 20%

B. 25%

D. 33.3%

D. 50%

Answer: B

Solution:

$$\text{Probability} = 1/\text{No. of Treatments Left Unpicked} = 1/(5-1) = 0.25 = 25\%$$

13. The return period of a flood stage of 5 meters or more is 2 years in a certain location. What is the probability that this event will be reached next year?

A. 2% B. 20% D. 50% D. 75%

Answer: C

Solution:

$$\text{Probability} = 1/\text{return Period} = 1/2 = 0.5 = 50\%$$

Difficult Questions

14. In Analysis of Variance, what is the F value of an experiment having treatment sum of squares of 2,510.5, error mean square error of 13.4, treatment degrees of freedom of 2 and error degrees of freedom of 12?

A. 15.61 B. 15.16 D. 93.86 D. 93.68

Answer: D

Solution:

$$\text{Tr MS} = \text{Tr SS} / \text{Tr DF} = 2510.5 / 2 = 1255.25$$

$$F = \text{Tr MS} / \text{MSE} = 1255.25 / 13.4 = 93.68$$

15. You need to present a graph of x values vs. y values. The x values are 0, 1, 2, 3, 4 and 5. The y values are ln 0, ln 1, ln 2, ln 3, ln 4 and ln 5, respectively. Since ln 0 results to a mathematical error, what value will you use for ln 0? Use 2 decimal places in your computations.

A. -6.91 B. -4.61 D. Infinity D. 6.91

Answer: B

Solution: Since infinity cannot be used for graphing and computation is limited to 2 decimal places, use ln 0.01 (closest to ln 0) instead of ln 0, then $\ln 0.01 = -4.61$

16. In a regression equation $Y = 2.5 + 3X$, where X is the flood stage in meters at a certain location and Y is the estimated amount in million Php of damaged crops downstream. If flood stage is 20 meters, what is the estimated amount of damaged crops in million Php?

A. 62.5 B. 65.2 D. 26.5 D. 25.6

Answer: A

Solution:

$$\begin{aligned} Y &= 2.5 + 3X \\ &= 2.5 + 3(20) \\ &= \text{Php}62.5 \text{ million} \end{aligned}$$